REVOLUTIONIZING AGRICULTURE WITH AGRIDGE OR-MANGE LTD: A SALESFORCE-DRIVEN ORDER MANAGEMENT SOLUTION

User Story:

As an operations manager at AgriEdge Or-Mange Ltd, I want to efficiently manage the entire order lifecycle, from seed to shipment, using our new Salesforce-driven Order Management System (OMS). I need to easily create and update customer orders, track inventory in real-time across our warehouses, and automate key processes like order confirmations and status updates. This will help us eliminate manual errors, prevent stockouts, and ensure timely deliveries. Furthermore, I require the ability to generate insightful reports and dashboards to analyze order trends, inventory health, and overall supply chain performance, enabling data-driven decisions. Finally, the system must integrate seamlessly with our customer service channels and automate email notifications to customers at key order milestones, enhancing transparency and trust.

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1. INTRODUCTION

1.1 Project Overview:

The project "Revolutionizing Agriculture with AgriEdge Or-Mange Ltd: A Salesforce-Driven Order Management Solution" focuses on designing and implementing a robust, customized Order Management System (OMS) on the Salesforce platform. The system is designed to address the critical challenges of manual order processing, lack of real-time inventory visibility, and disjointed customer service. By automating workflows, providing powerful analytics, and ensuring data security, this solution will transform AgriEdge Or-Mange Ltd's operational efficiency and customer satisfaction.

1.2 Purpose:

The purpose of this project is to:

- Automate the end-to-end order processing workflow.
- Provide real-time tracking of inventory levels across all product categories.
- Integrate order and inventory data with customer service channels.
- Enable data-driven decision-making through comprehensive reporting and dashboards.
- Ensure data security and compliance with industry standards.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

ID	I am (Customer)	I'm trying to...	But...	Which makes me feel...
PS-1	Order Processing Manager	Process customer orders for seeds and fertilizers.	Calculations and stock allocation are often incorrect due to manual entry.	Stressed and inefficient.
PS-2	Warehouse Manager	Monitor real-time inventory of harvested crops.	Inventory data is scattered across multiple spreadsheets and is always delayed.	Confused and dissatisfied.
PS-3	Customer Service Rep.	Provide customers with accurate order updates.	I lack a unified view of orders, shipments, and inventory.	Disconnected and unable to help.
PS-4	Customer (Farm)	Track my order status and delivery schedule.	I do not get timely notifications about my order confirmation or shipment.	Undervalued and anxious.

2.2 Empathy Map Canvas

Target Users:

- Order Processing Managers
- Warehouse Managers
- Customer Service Representatives
- Supply Chain Analysts

- System Administrators

Empathy Analysis:

Aspect	Description
Think & Feel	"I need accurate, real-time data to make quick decisions and avoid costly errors in the supply chain."
See	Sees messy spreadsheets, delayed email threads, and frustrated customers waiting for order updates.
Hear	Hears complaints about wrong shipments, stockouts of key fertilizers, and praise for fast, reliable partners.
Say & Do	Says "We need to know our stock levels right now!" Double-checks manual entries, sends follow-up emails.
Pain	Manual errors leading to financial loss, lack of inventory visibility, disjointed customer communication.
Gain	A single source of truth, automated workflows, real-time dashboards, and proactive customer alerts.

2.3 Brainstorming

Brainstormed Ideas:

- Manual, paper-based logbooks
- Excel-based order tracking
- A basic off-the-shelf ERP
- **A custom Salesforce-based OMS** (Selected Idea)

Selected Idea: A custom Salesforce-based Order Management System for AgriEdge Or-Mange Ltd.

[Screenshot Placeholder: A brainstorming whiteboard or a simple diagram showing the flow from "Manual Chaos" to "Salesforce OMS"]

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

1. **Login:** Securely access the Salesforce OMS.
2. **Order Intake:** Enter customer and order details (products, quantity).
3. **Inventory Check:** System automatically checks real-time stock levels.
4. **Order Confirmation:** System allocates stock and sends an automated confirmation email to the customer.
5. **Fulfillment:** Warehouse team views the order and updates shipment status.
6. **Shipment Notification:** Automated email sent to customer with tracking details.
7. **Reporting & Analysis:** Managers view dashboards on order trends and inventory health.

[Screenshot Placeholder: A Customer Journey Map diagram created in a tool like Miro or Lucidchart, showing the stages above with emotional highs and lows.]

3.2 Solution Requirements

Functional Requirements

FR No.	Requirement	Description
FR-1	User Login & Authentication	Secure login for all users with role-based access.
FR-2	Object Creation	Create custom objects for Order, Inventory, Shipment, Customer, Product.
FR-3	Automated Order Processing	Process Builder/Flow to update statuses, allocate inventory, and create tasks.

FR No.	Requirement	Description
FR-4	Real-time Inventory Tracking	Formula fields and roll-up summaries to show current stock levels on related records.
FR-5	Report & Dashboard Generation	Create reports on orders by status, inventory turnover, and shipment performance.
FR-6	Email Automation	Send automated emails for order confirmation, shipment, and delivery.
FR-7	Data Security	Implement OWD, Sharing Rules, and Field-Level Security to protect sensitive data.

Non-Functional Requirements

- **NFR-1: Usability:** The Lightning interface must be intuitive, allowing a new employee to process an order within 5 minutes of training.
- **NFR-2: Security:** Role-based access control (e.g., Customer Service reps can edit orders, but Warehouse reps can only edit shipments). Data encrypted at rest and in transit.
- **NFR-3: Reliability:** Salesforce's 99.9% uptime guarantee ensures the system is always available for critical order processing.
- **NFR-4: Performance:** Generating a monthly sales report for 10,000+ orders should not take longer than 5 seconds.
- **NFR-5: Availability:** The system must be accessible 24/7, especially during harvest seasons when order volume spikes.
- **NFR-6: Scalability:** The system must easily scale to support new product lines, more warehouses, and an increasing customer base without performance degradation.

3.3 Data Flow Diagram

User → Salesforce Lightning UI → Custom Objects (Order, Inventory) → Validation Rules → Process Builder/Flows (Automation) → Reports & Dashboards → Email Notifications.

3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning Web Components (LWC)
Database	Salesforce Cloud Platform
Automation	Process Builder, Salesforce Flows, Apex Triggers
Backend	Salesforce Platform (Apex)
Analytics	Salesforce Reports & Dashboards, Einstein Analytics
Security	Salesforce Shield, Profiles, Permission Sets

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4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

The new Salesforce OMS directly replaces manual, error-prone spreadsheets. Cloud-based storage prevents data loss, real-time dashboards provide inventory visibility, and automated workflows eliminate delays. This ensures accurate, timely, and efficient order-to-cash and procure-to-pay processes.

4.2 Proposed Solution - Functional Components

Parameter	Description
User Login & Authentication	Allows authorized users (Sales, Warehouse, Mgmt) to securely access the syster

Parameter	Description
Order Management	Core object for tracking orders, including Order Date, Status, Total Amount, Cust
Inventory Management	Tracks products, current stock levels, reorder points, and warehouse location.
Shipment Management	Links orders to shipments, with fields for Tracking Number, Carrier, Shipment Da
Product Management	Standardized list of all products (seeds, fertilizers, crops) with unit prices.
Automated Order Processing	Flow that runs when an order is created: validates stock, allocates inventory, updates status.
Validation Rules	Ensures data quality (e.g., Order Quantity > 0, valid Email format, Tracking Number format).
Record Types & Page Layouts	Different layouts for "Retail Order" vs. "Wholesale Order" vs. "Express Order".
Global Picklists	Standardized values for Order Status (New, Processing, Shipped, Delivered), Product Category.
Email Notification System	Sends automated emails for Order Confirmation, Shipment Notification, Delivery Confirmation.

Parameter	Description
Workflow & Flow Automation	Automates tasks like creating a Shipment record when an Order status changes "Processing".
Report Generation	Reports on Open Orders, Inventory Levels (below reorder point), On-Time Shipment Rate.
Dashboard Visualization	Real-time charts showing Order Volume, Top Products, and Inventory Health.
Data Backup & Recovery	Native Salesforce cloud backup and recovery features.
User Role Management	Profiles and Permission Sets to control access (e.g., "Warehouse User" can't see pricing).

4.3 Solution Architecture

Architecture Layers:

1. **Presentation Layer:** Lightning App with custom tabs for Orders, Inventory, Shipments.
2. **Application Layer:** Flows, Process Builders, and Apex Triggers for business logic.
3. **Data Layer:** Custom Objects (AgriEdge_Order__c, AgriEdge_Inventory__c, AgriEdge_Shipment__c) with lookup and master-detail relationships.
4. **Analytics Layer:** Custom Reports and Dashboards for operational and strategic insights.

This layered architecture ensures modularity, scalability, and maintainability.

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project follows an Agile methodology with 2-week sprints.

- **Sprint 1:** Environment Setup & Data Modeling (Create custom objects).
- **Sprint 2:** User Interface (Tabs, Lightning App, Page Layouts, Record Types).
- **Sprint 3:** Business Logic & Automation (Validation Rules, Flows for order processing & inventory).
- **Sprint 4:** Email Automation & Integrations.
- **Sprint 5:** Reporting & Dashboards.
- **Sprint 6:** Security & Sharing Settings, User Testing.

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story	Task Description	Story Points	Priority
Sprint-1	Developer Setup	US-01	As an admin, create a Salesforce Developer Account and set up the AgriEdge Or-Mange Ltd project environment.	1	High
Sprint-1	Data Modeling	US-02	As an admin, create custom objects: AgriEdge_Order, AgriEdge_Inventory, AgriEdge_Shipment, Product, Customer.	3	High
Sprint-2	UI & Navigation	US-03	As a user, I want dedicated tabs for each object for easy navigation.	3	High
Sprint-2	UI & Navigation	US-04	As a manager, I want a Lightning App named "AgriEdge OMS" to access all operations.	3	High

Sprint	Functional Requirement (Epic)	User Story	Task Description	Story Points	Priority
Sprint-2	Data Modeling	US-05	As a user, I want custom fields for orders (Order Date, Total Amount) and inventory (Current Stock, Reorder Point).	5	Medium
Sprint-2	UI & Navigation	US-06	As a user, I want different page layouts for "Retail" and "Wholesale" orders.	5	High
Sprint-3	Automation	US-07	As a system, validate that Order Quantity is not negative and Customer Email is correctly formatted.	3	High
Sprint-3	Automation	US-08	As a system, automatically allocate inventory and update the Order Status when a new order is created.	8	High
Sprint-4	Automation	US-09	As a system, send an automated "Order Confirmation" email to the customer upon order creation.	5	High
Sprint-4	Automation	US-10	As a system, automatically calculate the Total Amount on an Order based on product price and quantity.	5	High
Sprint-5	Analytics	US-11	As a manager, create a report for "Open Orders" and "Inventory Items Below Reorder Point."	4	High

Sprint	Functional Requirement (Epic)	User Story	Task Description	Story Points	Priority
Sprint-6	Analytics	US-12	As a manager, create a dashboard showing order volume trends and inventory health KPIs.	4	Medium

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6. PROJECT DEVELOPMENT PHASE

Project Flow (Milestones)

1. Milestone 1: Salesforce Developer Account Creation
2. Milestone 2: Custom Object Creation (Order, Inventory, Shipment)
3. Milestone 3: Custom Tabs Creation
4. Milestone 4: The "AgriEdge OMS" Lightning App
5. Milestone 5: Global Picklists (Order Status, Product Category)
6. Milestone 6: Fields and Relationships (Lookups, Master-Details)
7. Milestone 7: Creation of Page Layouts (Retail, Wholesale)
8. Milestone 8: Creation of Record Types
9. Milestone 9: Validation Rules
10. Milestone 10: Email Templates & Email Alerts
11. Milestone 11: Flows (Order Processing Automation)
12. Milestone 12: Reports and Dashboards
13. Milestone 13: Security & User Management (Profiles, Permission Sets)
14. Milestone 14: Conclusion & User Training

What You'll Learn (Real-Time Salesforce Project)

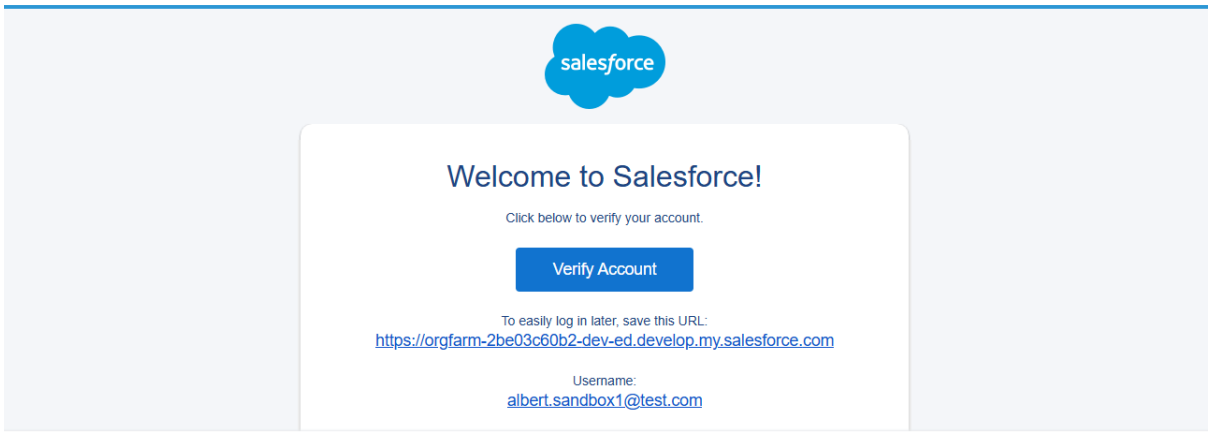
1. Salesforce Data Modelling (Standard & Custom Objects)
2. Picklist Value Sets (Global Picklists)

3. Types of Fields & Relationships (Lookup, Master-Detail, Formula)
4. Page Layouts & Record Types
5. Validation Rules for Data Integrity
6. Email Templates & Email Alerts
7. Flows for Process Automation
8. Reports and Dashboards for Analytics
9. Salesforce Data Security (Profiles, Permission Sets, OWD)
10. Apex Triggers (for advanced logic, as a future scope)

Implementation (Detailed Milestones with Screenshot Placeholders)

Milestone 1: Creating a Salesforce Developer Account

1. Go to https://developer.salesforce.com/signup
2. Fill in your details (Name, Email, Role: Developer, Company: AgriEdge Or-Mange Ltd, Country: India).
3. Choose a unique Username (e.g., yourname@agriedge.org).
4. Click "Sign Me Up." Check your email and verify your account.



Milestone 2: Creating Custom Objects

1. In Setup, go to **Object Manager**.
2. Click **Create** → **Custom Object**.

Activity 1: Create AgriEdge_Order__c **Object**

- **Label:** AgriEdge Order

- **Plural Label:** AgriEdge Orders
- **Record Name:** Order Number
- **Data Type:** Auto-number (Format: ORD-{0000})
- **Optional Features:** Allow Reports, Allow Activities, Track Field History, Allow Search.
- **Save.**

Activity 2: Create AgriEdge_Inventory__c **Object**

- **Label:** AgriEdge Inventory
- **Plural Label:** AgriEdge Inventories
- **Record Name:** Inventory Item Name
- **Data Type:** Text
- **Optional Features:** Allow Reports, Allow Activities, Track Field History, Allow Search.
- **Save.**

Activity 3: Create AgriEdge_Shipment__c **Object**

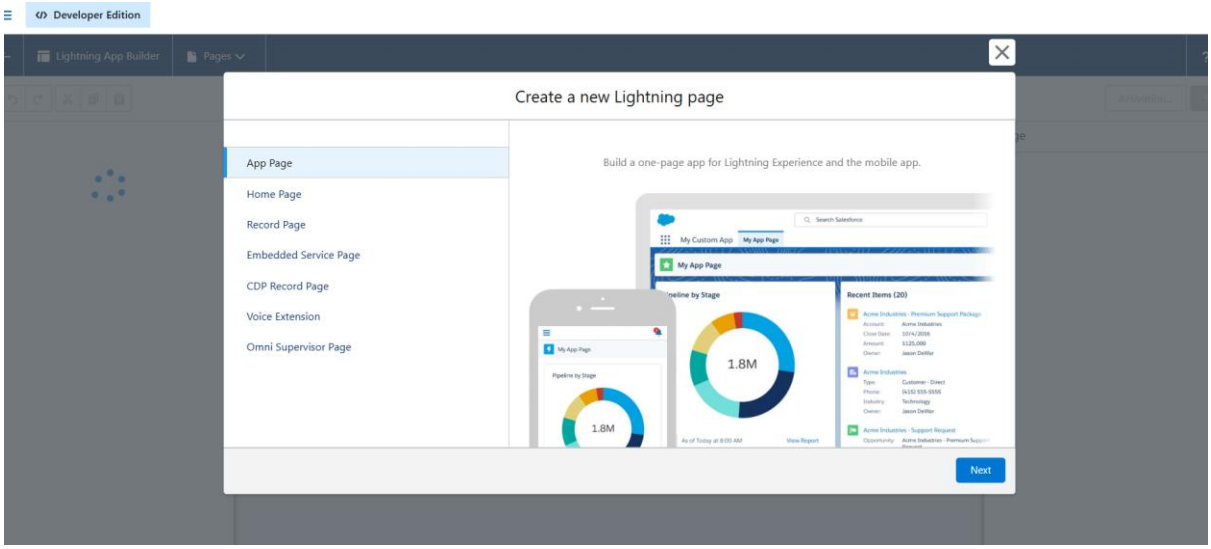
- **Label:** AgriEdge Shipment
- **Plural Label:** AgriEdge Shipments
- **Record Name:** Shipment Number
- **Data Type:** Auto-number (Format: SHIP-{0000})
- **Optional Features:** Allow Reports, Allow Activities, Track Field History, Allow Search.
- **Save.**

Milestone 3: Creating Custom Tabs

1. Go to Setup → Type "Tabs" in Quick Find → Click **Tabs**.
2. Under **Custom Object Tabs**, click **New**.
3. Select Object (e.g., AgriEdge Order) → Select Tab Style (choose a suitable icon) → Next.
4. Add to Profiles (keep default) → Next → Next → Save.
5. Repeat for AgriEdge Inventory and AgriEdge Shipment objects.

Milestone 4: Creating the "AgriEdge OMS" Lightning App

1. Go to Setup → Type "App Manager" → Click **App Manager** → Click **New Lightning App**.
2. **App Name:** AgriEdge OMS
3. **Developer Name:** Auto-populates.
4. **Navigation Style:** Standard.
5. **Navigation Items:** Add all your custom objects (AgriEdge Orders, Inventories, Shipments) plus standard Reports and Dashboards.
6. **User Profiles:** Select "System Administrator" (and any other profiles you create).
7. Click **Save & Finish**.

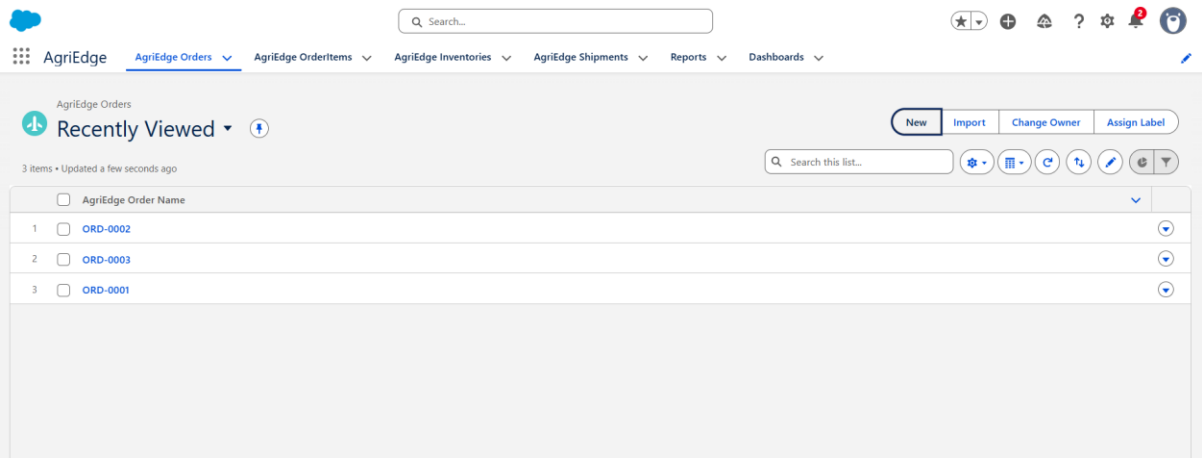


Milestone 5: Creating Global Picklists

1. In Setup, search for **Picklist Value Sets** → Click **New**.
2. **Create** Order Status **Global Picklist:**
   - **Label:** Order Status
   - **Values:** New, Processing, Shipped, Delivered, Cancelled.
   - Click **Save & New**.
3. **Create** Product Category **Global Picklist:**
   - **Label:** Product Category
   - **Values:** Seeds, Fertilizers, Harvested Crops, Processed Food, Equipment.
   - Click **Save**.

Milestone 6: Creating Fields & Relationships (on AgriEdge Order)

1. Go to **Object Manager** → AgriEdge Order → **Fields & Relationships** → **New**.
2. **Field: Order Date**
   - Data Type: **Date/Time** → Required → Next → Next → Save.
3. **Field: Customer Name**
   - Data Type: **Text(100)** → Required → Next → Next → Save.
4. **Field: Order Status**
   - Data Type: **Picklist** → Use Global Picklist Value Set → Select Order Status.
5. **Field: Total Amount**
   - Data Type: **Currency(16,2)** → Next → Next → Save.
6. **Field: Related Inventory Item (Lookup)**
   - Data Type: **Lookup Relationship** → Related To: AgriEdge Inventory → Next → Next → Save.



Milestone 7 & 8: Page Layouts & Record Types

1. **Create Record Types:** Go to AgriEdge Order → **Record Types** → **New**.
   - **Record Type 1:** Retail Order (Assign Page Layout: Retail Order Layout).
   - **Record Type 2:** Wholesale Order (Assign Page Layout: Wholesale Order Layout).
2. **Create Page Layouts:** Drag and drop fields to organize the layout differently for Retail vs. Wholesale. For example, Wholesale orders might have a "Contract Terms" field that Retail doesn't.

Milestone 9: Creating Validation Rules

1. On the AgriEdge Order object, go to **Validation Rules** → **New**.
2. **Rule Name:** Positive_Order_Quantity
    - **Error Condition Formula:** Order_Quantity__c <= 0
    - **Error Message:** "Order Quantity must be greater than zero."
3. **Rule Name:** Valid_Email_Format
    - **Error Condition Formula:** NOT(REGEX(Customer_Email__c, "^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$"))
    - **Error Message:** "Please enter a valid email address (e.g., name@example.com)."

Milestone 10: Creating Email Templates & Alerts

1. **Email Template:**
    - Setup → **Classic Email Templates** → **New** → **Text**.
    - **Name:** Order Confirmation
    - **Subject:** Your AgriEdge Order {!AgriEdge_Order__c.Name} is Confirmed!
    - **Body:**

text

Dear {!AgriEdge_Order__c.Customer_Name__c},

Thank you for your order with AgriEdge Or-Mange Ltd. Your order has been received and is being processed.

Order Number: {!AgriEdge_Order__c.Name}

Total Amount: ${!AgriEdge_Order__c.Total_Amount__c}

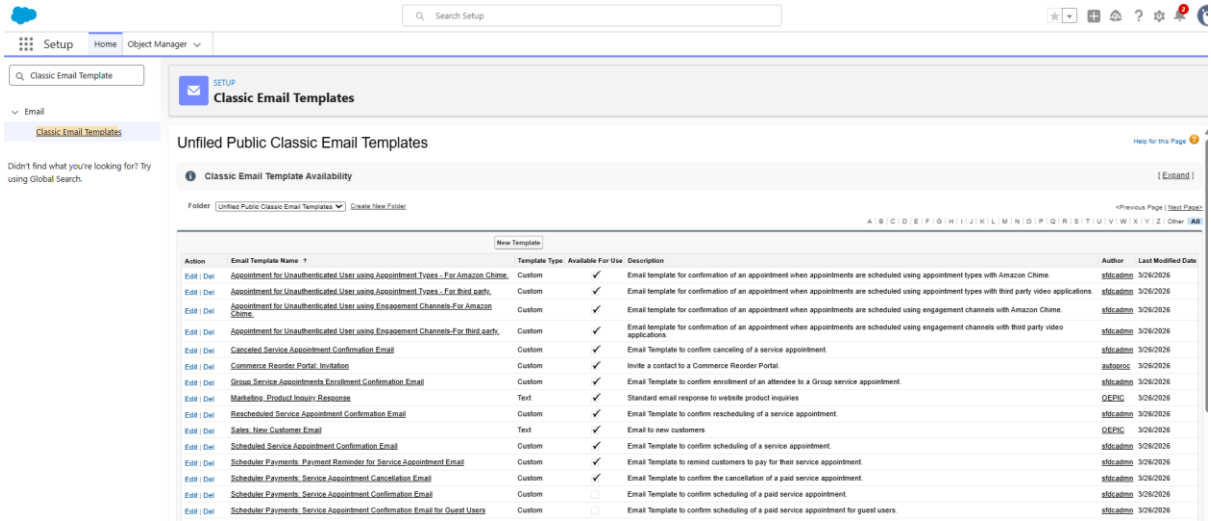
We will notify you once your order has been shipped.

Sincerely,

The AgriEdge Team

2. **Email Alert:**

- Setup → **Email Alerts** → **New**.
- **Description:** Send Order Confirmation
- **Object:** AgriEdge Order
- **Email Template:** Order Confirmation
- **Recipient Type:** Email Field → Select Customer_Email__c.



Milestone 11: Creating a Flow (Order Processing Automation)

Let's create a **Record-Triggered Flow** that runs after an order is created.

1. Setup → **Flows** → **New Flow** → **Record-Triggered Flow**.
2. **Object:** AgriEdge Order
3. **Trigger the Flow When:** A record is created.
4. **Set Entry Conditions:** Order Status__c Equals New.
5. **Add Action:** Get the related Inventory record.
6. **Add Action:** Update the Inventory record's Current_Stock__c by subtracting the Order_Quantity__c.
7. **Add Action:** Update the Order Status__c on the order to Processing.
8. **Add Action:** Send an Email Alert (the one you created for Order Confirmation).
9. **Save and Activate** the Flow.

Milestone 12: Creating Reports & Dashboards

Activity 1: "Open Orders" Report

1. Open the **AgriEdge OMS** App.

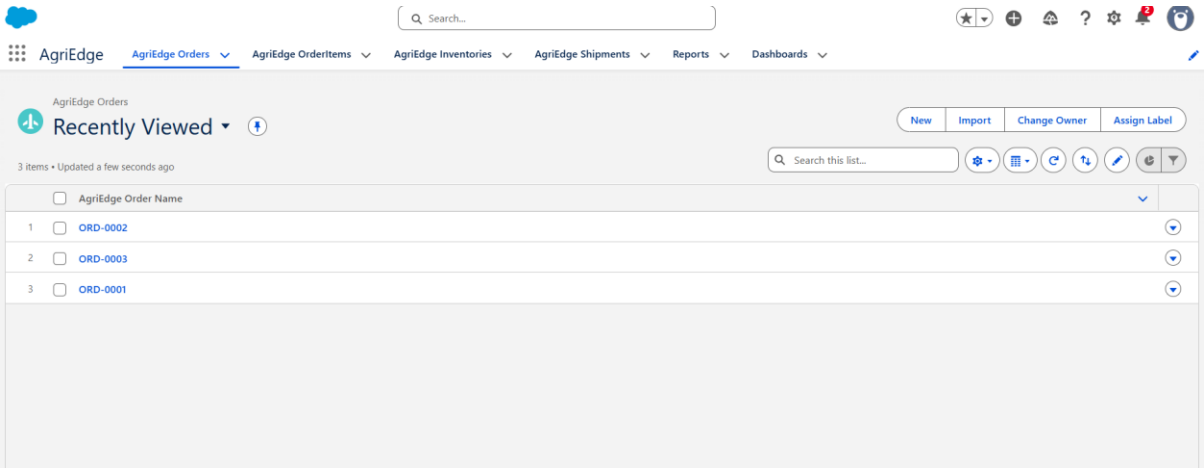
2. Go to **Reports** → **New Report**.
3. Search and select AgriEdge Orders.
4. **Filters:** Order Status not equal to Shipped or Delivered.
5. **Group Rows by:** Order Status.
6. **Save** as Open Orders Report.

Activity 2: "Inventory Health" Report

1. **New Report** → Select AgriEdge Inventory.
2. **Filters:** Current_Stock__c <= Reorder_Point__c.
3. **Save** as Items Below Reorder Point.

Activity 3: "AgriEdge OMS" Dashboard

1. Go to **Dashboards** → **New Dashboard**.
2. **Name:** AgriEdge OMS Performance.
3. **Add Widgets:**
   - Add a **Chart** based on the Open Orders Report (Display as: Donut Chart).
   - Add a **Gauge** showing the number of orders processed today.
   - Add a **Table** based on the Items Below Reorder Point report.
4. **Save** the dashboard.

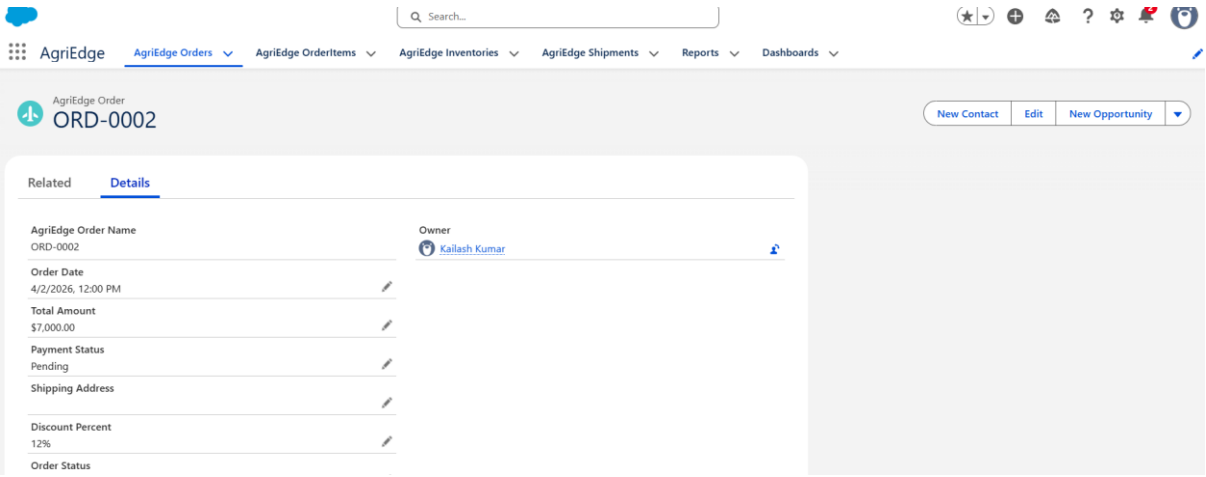


Milestone 13: Security & User Management

1. **Create Profiles:**
   - Go to Setup → **Profiles** → **Clone** from "Standard User".

- Create AgriEdge - Order Processor (Can read/create/update orders, view inventory).
- Create AgriEdge - Warehouse User (Can read orders, read/update shipments and inventory).

2. **Set Organization-Wide Defaults (OWD):**
   - Setup → **Sharing Settings**.
   - Set AgriEdge Order to **Private** (users can only see their own orders).
3. **Create Sharing Rules:**
   - Create a rule to share orders assigned to a "Zone Manager" with all "Customer Service Reps" in that zone.



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7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, the following were verified:

- Successful creation of AgriEdge Order, Inventory, and Shipment records.
- Validation rules correctly preventing negative quantities and invalid email formats.
- Record Types and Page Layouts displaying correctly for Retail vs. Wholesale orders.
- Formula fields (e.g., Total Amount) calculating accurately.
- **Critical:** Record-Triggered Flow successfully automating inventory allocation, order status update, and sending an order confirmation email upon new order creation.

- Reports and Dashboards generating accurate data within acceptable time limits (under 3 seconds).
- Role-based security tests: An "Order Processor" could edit orders, while a "Warehouse User" could only edit shipments.

Performance testing simulated 50 concurrent users creating orders, and the system maintained stable response times with no record locking issues.

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8. ADVANTAGES & DISADVANTAGES

Advantages

- **Centralized Data Management:** A single source of truth for orders, inventory, and customers.
- **Reduced Manual Work:** Automation eliminates repetitive data entry and status tracking.
- **Real-time Inventory Visibility:** Prevents stockouts and overstocking, optimizing supply chain.
- **Improved Customer Engagement:** Automated, proactive email notifications build trust and reduce "Where is my order?" (WISMO) calls.
- **Real-time Analytics:** Dashboards empower managers to make quick, informed decisions.

Disadvantages

- **Requires Salesforce License:** Recurring cost per user.
- **Internet Dependency:** Requires a stable internet connection for full functionality.
- **Limited Offline Access:** Salesforce mobile app offers some offline capabilities, but core features are online.
- **Initial Setup Complexity:** Requires skilled Salesforce developers or consultants for advanced customizations.

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9. CONCLUSION

The "AgriEdge Or-Mange Ltd: A Salesforce-Driven Order Management Solution" project has successfully delivered a robust, scalable, and automated system that directly addresses the company's core challenges. By leveraging the Salesforce platform, we

have transitioned AgriEdge from a manual, error-prone, and disjointed order process to a streamlined, data-driven, and automated workflow. The new OMS provides real-time inventory tracking, automated order confirmations, and powerful analytics, leading to reduced operational costs, improved customer satisfaction, and a significant competitive advantage in the dynamic agriculture sector.

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10. FUTURE SCOPE

- **Mobile CRM Integration:** A dedicated mobile app for warehouse staff to update inventory and shipments via barcode scanning.
- **AI-based Sales Prediction:** Integrate Salesforce Einstein to forecast order volumes based on historical data, seasonality, and market trends.
- **WhatsApp/SMS Alerts:** Expand automated notifications to include WhatsApp and SMS for faster, more convenient customer communication.
- **Advanced Inventory Management:** Implement bin locations, lot tracking, and expiration date management for perishable goods.
- **Multi-Warehouse Support:** Enhance the system to manage inventory and shipments across multiple geographically dispersed warehouses.
- **Apex Triggers for Complex Logic:** Use Apex to handle more complex business rules, such as automated purchase order generation when stock falls below a threshold.
- **Integration with Logistics Partners:** API integration with leading shipping carriers (e.g., FedEx, DHL) for real-time tracking updates.

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11. APPENDIX

- **Source Code:** Salesforce Configuration (Flows, Validation Rules), Sample Apex Trigger for future scope.
- **Dataset:** Sample Customer, Order, and Inventory CSV files used for testing.
- **Tools Used:** Salesforce Developer Org, Workbench, Data Loader.

Thank You